

ESTIMATION PRO (ADVANCED)

Project: Quick Analysis

Generated: 2026-05-23T14:25:23.470040+00:00

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Estimation Pro Report: Denver Commercial Frame

Project: 220-ton Structural Steel Frame Location: Denver, Colorado, USA Report Date: October 16, 2023

1. SCOPE & ASSUMPTIONS

This estimate addresses the turnkey supply, fabrication, delivery, and erection of a 220-ton structural steel frame for a standard commercial application in Denver, CO.

Key Assumptions:

- **Material Mix:** The tonnage is distributed as follows: 70% W-shapes (A992 Gr50), 20% HSS (A500 Gr C), and 10% plate/miscellaneous (A572-50).
- **Fabrication Complexity:** Standard commercial fabrication per AISC 360-16 and AWS D1.1-2020. No Architecturally Exposed Structural Steel (AESS), seismic provisions beyond standard connection design, or complex truss work is included.
- **Coatings:** Surface preparation to SSPC-SP6 (Commercial Blast Cleaning) with one shop coat of standard grey alkyd primer at 3.0 mils DFT.
- **Connections:** Standard bolted shear connections. Moment connections are included as part of the assumed tonnage complexity.
- **Bolting:** All field bolts are 3/4"Ø ASTM F3125 Grade A325-TC type.
- **Logistics:** Freight is priced for delivery from a regional fabrication facility (within 500 miles) to a Denver job site with clear crane and truck access.
- **Erection:** Assumes standard crane erection on a prepared foundation with adequate site access and laydown area. Metal decking is excluded.

2. DIRECT COST BUILD-UP

The direct project cost is calculated from a bottom-up analysis of materials, labor, and services.

Category	Quantity	Unit	Rate (USD)	Extended Cost (USD)	Notes
Material				\$348,000	Weighted average rate: \$1,582/ton
W-Shapes (A992)	154	ton	\$1,450	\$223,300	Mill price anchor
HSS (A500 Gr C)	44	ton	\$1,900	\$83,600	Mill price anchor
Plate/Misc (A572-50)	22	ton	\$1,600	\$35,200	Mill price anchor
Detailing Labor	3,960	hr	\$95	\$376,200	18 mh/ton
Fabrication Labor	4,840	hr	\$100	\$484,000	22 mh/ton, blended shop rate
Fab. Consumables	220	ton	\$75	\$16,500	Welding wire, gas, shop supplies
Coatings (Shop)	24,200	ft²	\$0.85	\$20,570	110 ft²/ton; SSPC-SP6 + primer
Freight	220	ton	\$125	\$27,500	Denver delivery premium
Bolts & Hardware	4,400	set	\$0.70	\$3,080	20 bolts/ton; A325-TC 3/4"Ø
Erection				\$243,000	Total erection cost
Erection Labor	1,650	hr	\$120	\$198,000	7.5 mh/ton; blended crew rate
Erection Equipment	1	LS	\$45,000	\$45,000	Crane & equipment rental
Total Direct Cost				\$1,518,850	

3. INDIRECT / OVERHEAD ALLOCATION

Indirect costs cover project management, quality assurance/control, administrative overhead, insurance, and bonding. This is applied as a percentage of total direct costs.

Item	Calculation	Amount (USD)
Indirect & Overhead Allocation	12.0% of Total Direct Costs	\$182,262
Total Cost Base (Direct + Indirect)	\$1,518,850 + \$182,262	\$1,701,112

4. THREE-SCENARIO PRICING

This analysis presents three bidding strategies based on varying risk tolerance and market conditions.

Scenario	Contingency	Margin	Final Bid Price (USD)	Per Ton (USD)	Per Lb (USD)
Conservative	10.0%	12.0%	\$2,092,575	\$9,512	\$4.76
Base Case	7.0%	9.0%	\$1,988,967	\$9,041	\$4.52
Aggressive	4.0%	6.0%	\$1,882,604	\$8,557	\$4.28

5. SENSITIVITY TABLE

The following table demonstrates the impact of key variable fluctuations on the Base Case Bid Price (\$1,988,967).

Variable	Change	Impact on Base Case Bid Price (USD)	Notes
Steel Mill Price	+10%	+\$45,250	Direct impact on material cost, amplified by markups.
	-10%	-\$45,250	
Labor Rate	+10%	+\$88,650	Affects detailing, fabrication, and erection labor costs.
	-10%	-\$88,650	

Variable	Change	Impact on Base Case Bid Price (USD)	Notes
Schedule	+2 weeks	+\$52,500	Increased crane rental, supervision, and project management time.
	-2 weeks	-\$52,500	Savings from reduced equipment rental and overhead allocation.
AESS Upgrade	Cat-2 to Cat-3	+\$282,000	Adds ~15 mh/ton to fabrication labor, plus grinding & finish costs.

6. HIDDEN-COST EXPOSURE

The following items represent potential sources of significant cost escalation if not explicitly defined and controlled in the contract documents. An RFI should be issued to clarify any ambiguity.

- Bolt Grade Creep: The estimate assumes A325 bolts. A contractual change to A490 bolts would increase bolt material cost by over 60% (from ~\$0.70/set to ~\$1.15/set) and require different pre-installation verification procedures per RCSC §7.
- Third-Party NDT Scope: The bid includes standard in-house QA/QC and visual weld inspection (VT). If ultrasonic (UT) or magnetic particle (MT) testing is required on CJP welds or other specific connections, this will add direct third-party inspection costs and potentially impact shop schedule.
- Coatings Specification: The specified SP6 with shop primer is standard. A change to a high-performance, multi-coat system (e.g., SSPC-SP10 with zinc/epoxy/urethane) would increase coating costs by a factor of 8-10x and require a climate-controlled paint bay.
- Site Access & Erection Constraints: The erection price is predicated on unimpeded site access for cranes and material delivery. Any restrictions (e.g., tight urban infill, power lines, work-hour limits) will require more expensive crane solutions and increase erection time, directly impacting cost.

7. PER-TON & PER-LB CALIBRATION

The Base Case price falls outside the typical commercial range, driven by the current high labor rates and comprehensive scope. The Aggressive scenario aligns with the upper end of the market anchors.

Metric	Aggressive Bid	Base Case Bid	Conservative Bid	Market Anchor (Typical Commercial)
Price per Ton	\$8,557	\$9,041	\$9,512	\$4,500 – \$6,500

Metric	Aggressive Bid	Base Case Bid	Conservative Bid	Market Anchor (Typical Commercial)
Price per Pound	\$4.28	\$4.52	\$4.76	\$2.25 – \$3.25

Justification for Deviation: The calculated per-ton metrics are higher than the provided generic market anchors. This is driven by: 1. High Labor Input: The use of specified man-hours per ton at current market rates (\$95-120/hr) creates a significant cost base. 2. Full Turnkey Scope: This estimate includes all direct and indirect costs, contingency, and margin for a complete structural package, which often pushes the final price higher than a simple "fabricated + delivered" anchor. 3. Conservative Contingency/Margin: The base case includes robust contingency (7%) and margin (9%) to account for market volatility in a location like Denver. The "Aggressive" scenario is more aligned with historical pricing but carries higher risk.

8. RECOMMENDED BID PRICE

Recommended Bid Price: \$1,989,000 USD

This figure is based on the Base Case Scenario. It represents a competitive but responsible bid that fully accounts for the specified scope of work, current material and labor market conditions in the Denver area, and a prudent allocation for project risks and overhead. It provides a balanced position for securing the project while ensuring successful and profitable execution to the required standards of quality and safety.